MSG No. 147.15 $P\bar{3}'$ [Type III, trigonal]

Table 1: Wyckoff site: `1a`, site symmetry: `-3'..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,5,6]`

Table 2: Wyckoff site: `1b`, site symmetry: `-3'..`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,3,4,5,6]`

Table 3: Wyckoff site: `2c`, site symmetry: `3..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3]`
2	$[0,\ 0,\ -z]$	`[4,5,6]`

Table 4: Wyckoff site: `2d`, site symmetry: `3..`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,2,3]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[4,5,6]`

Table 5: Wyckoff site: `3e`, site symmetry: `-1'`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,4]`
2	$\left[0,\ \frac{1}{2},\ 0\right]$	`[2,5]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[3,6]`

Table 6: Wyckoff site: `3f`, site symmetry: `-1'`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[1,4]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,5]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,6]`

Table 7: Wyckoff site: `6g`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-y,\ x-y,\ z]$	`[2]`

continued ...

Table 7

No.	position	mapping
3	$[-x+y,\ -x,\ z]$	[3]
4	$[-x,\ -y,\ -z]$	[4]
5	$[y,\ -x+y,\ -z]$	[5]
6	$[x-y,\ x,\ -z]$	[6]